Data Structures

Sample educational document for testing a PDF question-answering chatbot.

1. Array

An array stores elements in indexed positions. Accessing an element by index is typically constant time, while inserting into the middle may require shifting elements.

2. Linked list

A linked list consists of nodes that contain data and links to other nodes. Inserting after a known node can be efficient, but accessing an arbitrary position requires traversal.

3. Stack and queue

A stack follows Last In, First Out (LIFO). A queue follows First In, First Out (FIFO). Stacks are used for function calls and undo operations; queues are used in scheduling and buffering.

4. Hash table

A hash table uses a hash function to map keys to storage locations. Collisions occur when different keys map to the same location and can be handled using chaining or open addressing.

5. Tree

A tree is a hierarchical data structure made of nodes and edges. In a binary search tree, values in the left subtree are smaller than the node value and values in the right subtree are larger, under the usual unique-key convention.

Note: This is a short synthetic sample created for software testing, not an official textbook.